

CSY2084 Report

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Server Administration and Security

CSY2085

Month:01/03/2025

By: Olasubomi Rotimi

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Introduction

In this report, I will aim to show setting up, configuring and implementing security inside windows and Linux environment. As the role I am is a system administrator for a startup, my role will be to create an operational and secure infrastructure able to run online application and databases.

In this report, I will be showing how the setup, configuration and security can be implemented within both Windows and Linux server environments. As a system administration for a growing tech start up, my role is to establish an operational and secure infrastructure which is fully able to support online application and databases. This task will require full functionality and secure security measures to protect sensitive data transactions.

Objectives

The main objective of this project is:

- To ensure all server environments are fully functional
- Host web applications and databases with no errors and checks all requirements
- Secure server environment with effective security, implementing backups and recovery measure
- Document all configurations and security added in an industry standard form
- Reflect with insightful explanations on challenges faced

Planned Approach:

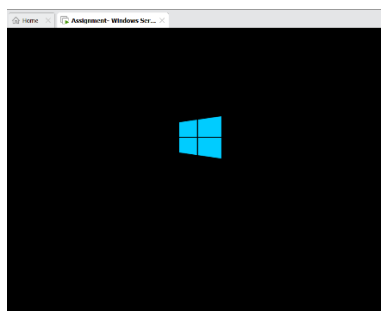
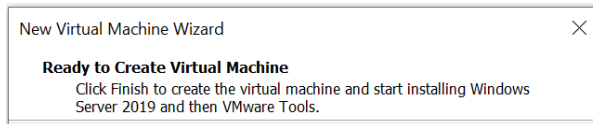
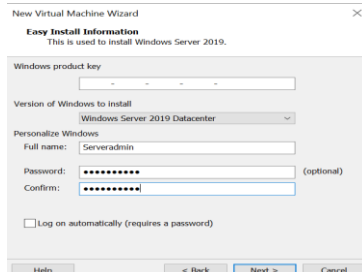
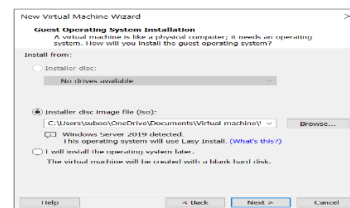
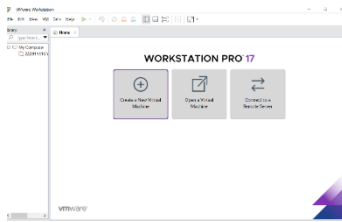
My plan for this project is to use on-premises servers not a cloud-based server. I will also be using VMware workstation which will give me the flexibility I need when testing and be within my control. As this plan allows for full control of the server setting, networking and security I won't need any third party cloud platforms in turn reducing security risk.

Task 1: Practical Implementation(40%)

Windows Server Environment Setup

Installation & Configuration

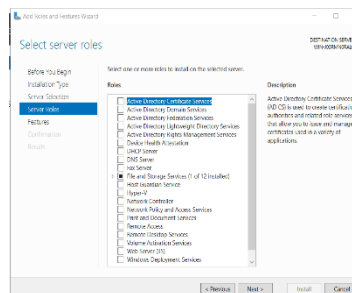
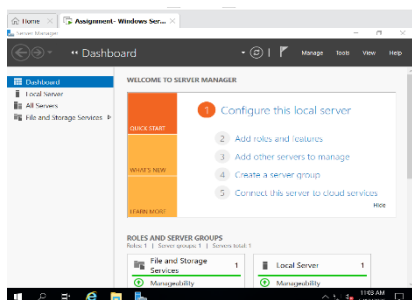
To start the assignment, I configured a virtual machine using VMware workstation to host windows server 2022. The VM was configured with 2 virtual CPUs, 4GB of RAM and a 40 GB virtual hard disk, I chose these specifications as they would ensure a stable environment when running later tasks. during the creation process I mounted the windows server 2022 ISO for the boot disk and carried on the setup instructions until the operating system was installed

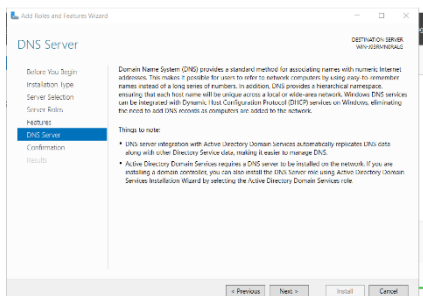


Next the configuration process involved including the correct time and default administrator account. Once the installation was complete, I renamed the computer and connected it to a NAT virtual network to allow for internet access which would later help with downloads and updates.

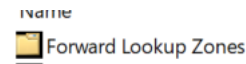
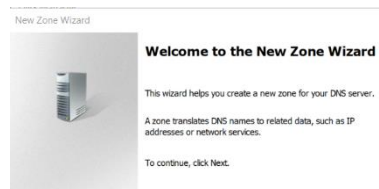
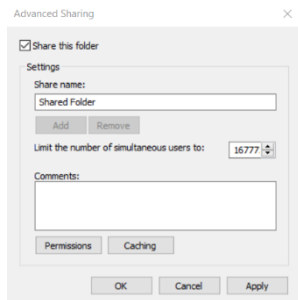
DNS & File Services

After completing the initial OS installation, I accessed Server Manager and went to the process to install and configure the DNS server and File and Storage Services roles. The DNS Server role is there to “translate requests for names into IP address”[1] the role DNS server play is essential when managing systems or hosting websites with specific domains.





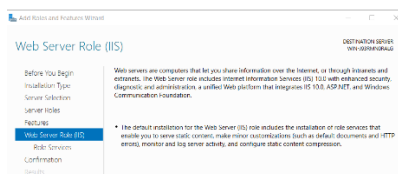
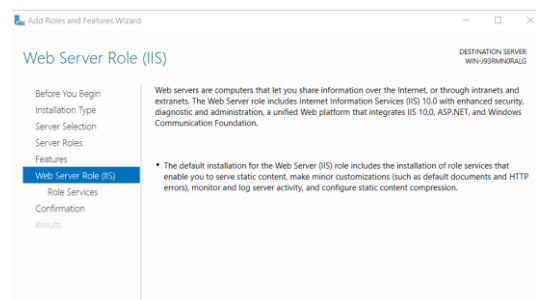
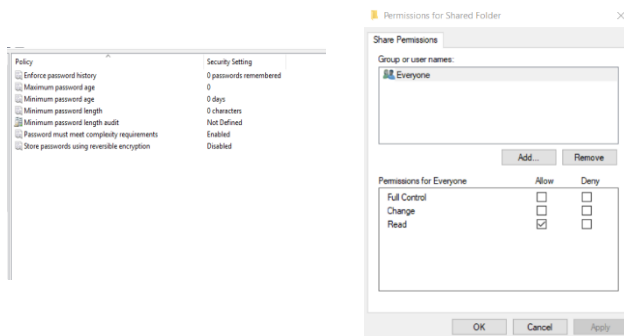
Once the role was installed, I created a Forward Lookup zone which was under the DNS Manager. Forward Lookup zone is used to change domain names into IP addresses. To test it was working properly I tested it using nslookup command



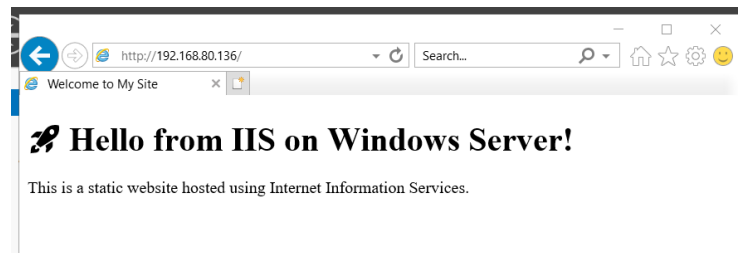
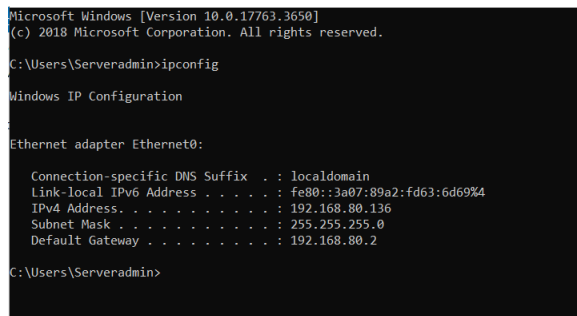
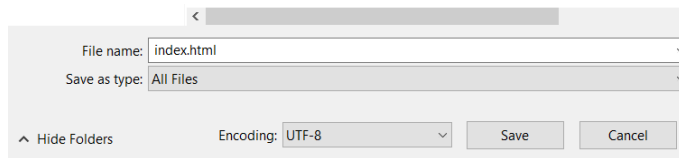
File sharing was created by making a folder called SharedFiles on the C: drive and then shared on the network. I edited the permission and set it so it can be read by everyone but not edited or deleted by everyone. This shows controlled access within the file sever

IIS and Hosting Static Site

IIS (Internet Information Services) is used to “enables Web applications to benefit from functionality” [2]To use it I had to install the role from Server Manager . When installing I selected certain options which would be essential for later use and removed some that would not be.

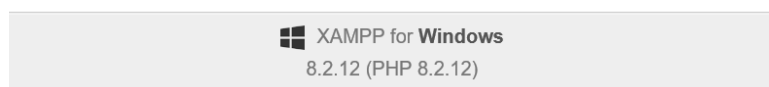
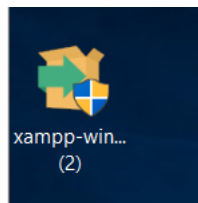


Once the installation was completed, I accessed the IIS root directory(C:\inetpub\wwwroot). Within this file I created a HTML file called index.html with a customised message. By using the server IP address, I accessed this file on the host machine which loaded the index.html file successfully which showed that IIS was working.

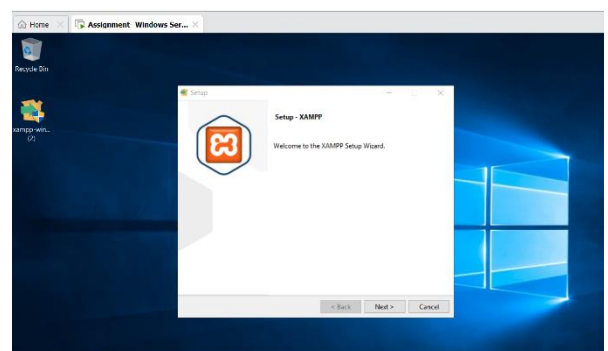
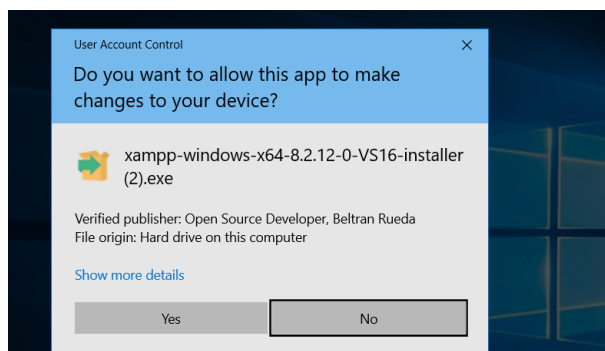


XAMPP Web App & MySQL

Dynamic content as well as database driven application often needs to use tools like XAMPP stack as it includes more tools like Apache, PHP and MySQL and are preconfigured saving time and speed up the process for web applications.



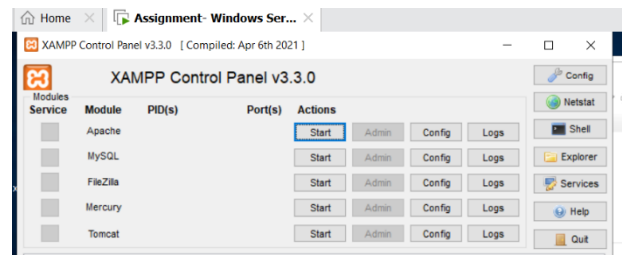
To install I copied the download file from the host into the VM once the installation was completed, I copied the PHP app into the XAMPP, htdocs directory and then created a SQL database using PhpMyAdmin which is also included in XAMPP. This is used to connect the database (student dB) to the browser using the correct credentials. While attempting this I ran into some problems due to not having the correct credentials to mitigate this problem I changed the password and was displayed a message to say I was grateful. Showing us, the database is reachable.



```
dbconnect.php - Notepad
File Edit Format View Help
<?php
$servername = "localhost";
$username = "root";
$password = "";
$dbname = "webapp_db";

// Create connection
$conn = new mysqli($servername, $username, $password, $dbname);

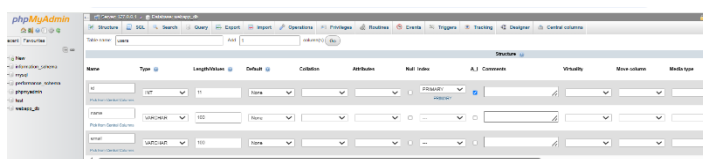
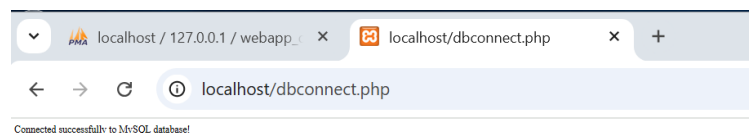
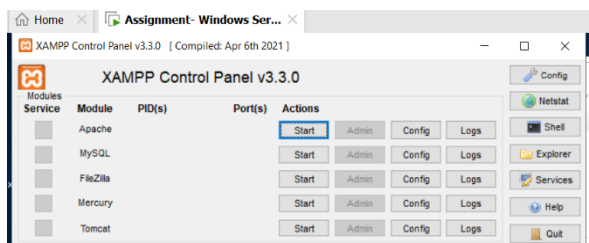
// Check connection
if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}
echo "Connected successfully to MySQL database!";
?>
```



```
dbconnect.php - Notepad
File Edit Format View Help
<?php
$servername = "localhost";
$username = "root";
$password = "";
$dbname = "webapp_db";

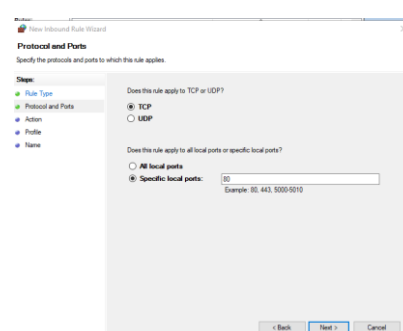
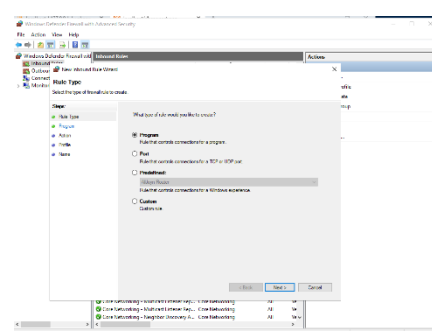
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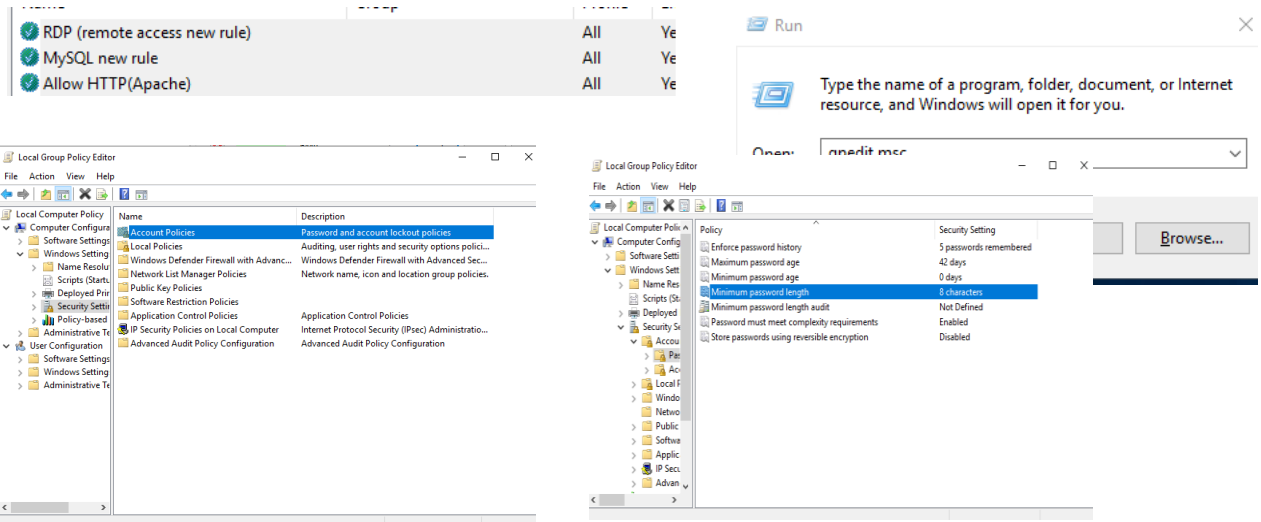
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echo "Connected successfully to MySQL database!";
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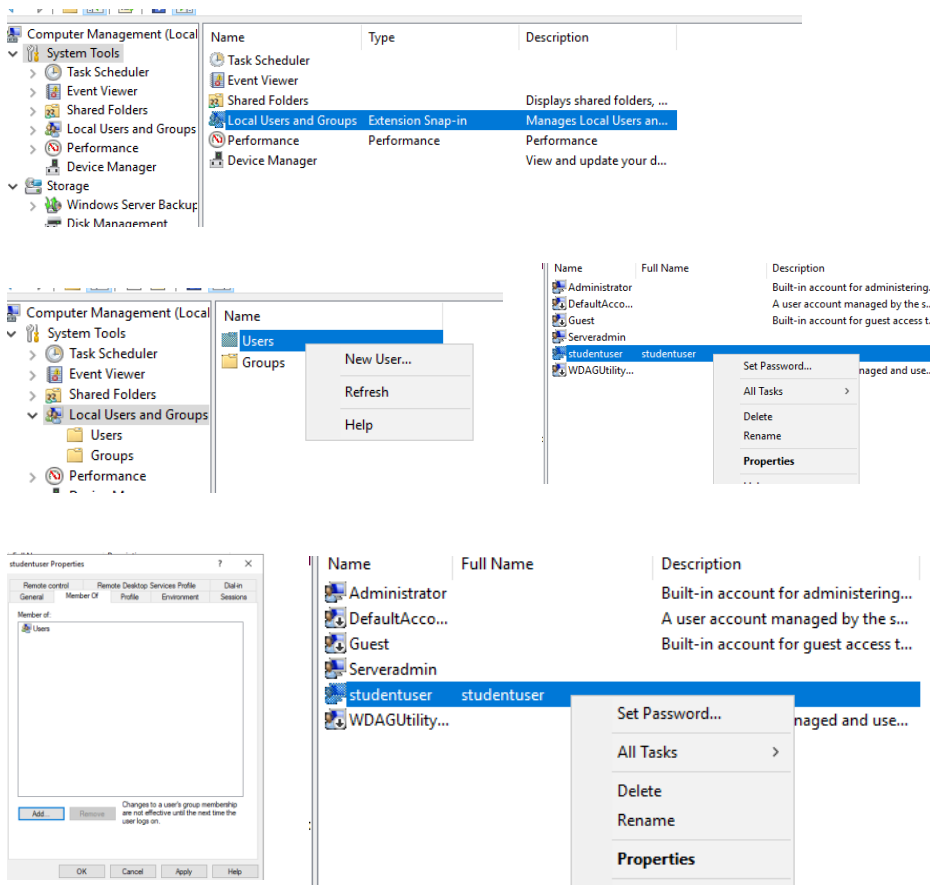
Windows Server Security

To secure the operating system I used windows defender firewall, group policy management console and BitLocker encryption. Firstly, I configured Windows defender firewall to stop any unsecure services working, leaving only the important services such HTTP Port 80, HTTPS port 443 and MySQL – 3306

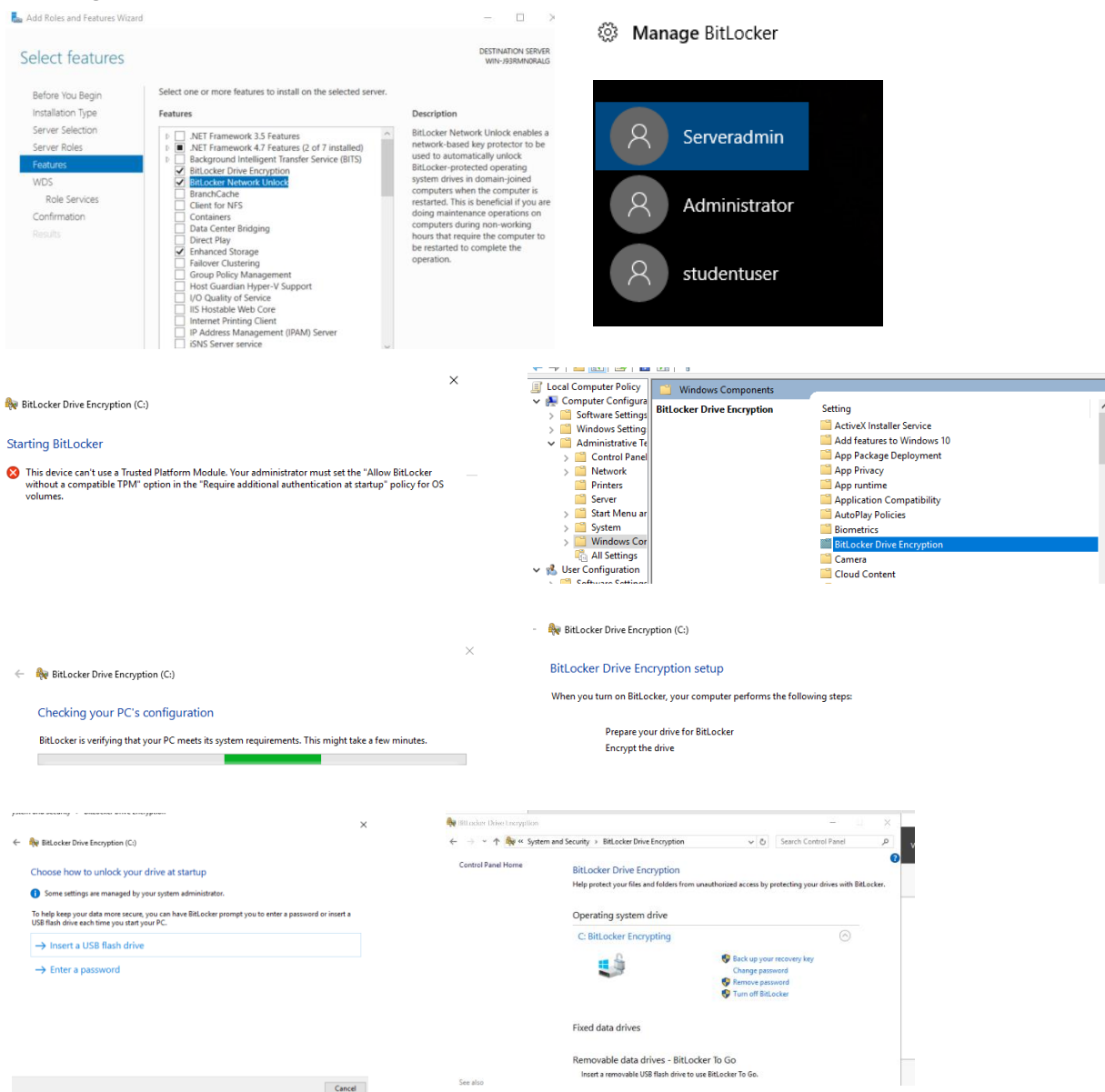




Next for the GPMC the main idea was for this to handle password policies. While doing this I decided to create a user account with less privileges and only for general use which would leave the main account for sensitive administrative task



Finally, I enable BitLocker on the C: drive by enabling this I was able to utilise a [3] “Windows security feature that provides encryption for entire volumes” Included in the process of enabling I saved the recovery key in safe external location and encryption in case attacks that involves stealing.



Linux Server Environment Setup

Installing Ubuntu Server

For creating the Linux environment, I used ubuntu server as it's known for hosting web servers. On VMware Workstation, the virtual machine I created was with 2 CPUs, 2gm ram and 20GB of disk space, after installing the ISO from online on my host machine I copied it not the VM and installed Ubuntu from the ISO, creating a user account during the setup. Once completed I logged in with the earlier user credentials and used the terminal how to check the IPP address by running the command "Ip a".

```
ubuntuusername@ubuntuservername:~$ ping 8.8.8.8 -c 4
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=128 time=116 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=128 time=10.6 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=128 time=11.0 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=128 time=11.5 ms

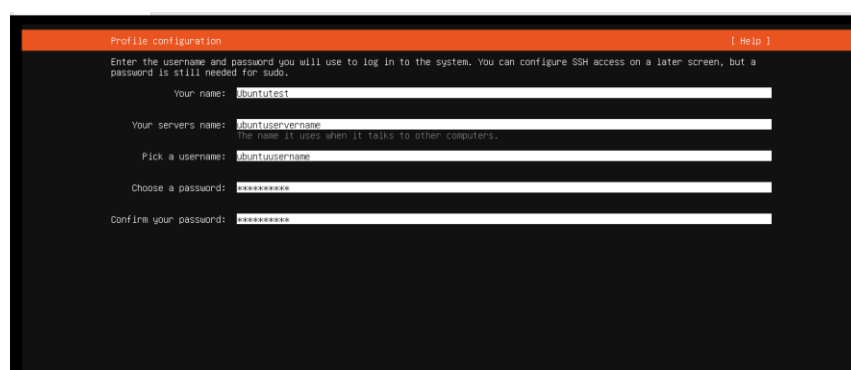
--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3007ms
rtt min/avg/max/mdev = 10.616/37.256/115.919/45.417 ms
ubuntuusername@ubuntuservername:~$ ping google.com -c 4
PING google.com (142.250.178.14) 56(84) bytes of data.
64 bytes from lhr48s27-in-f14.1e100.net (142.250.178.14): icmp_seq=1 ttl=128 time=8.75 ms
64 bytes from lhr48s27-in-f14.1e100.net (142.250.178.14): icmp_seq=2 ttl=128 time=8.26 ms
64 bytes from lhr48s27-in-f14.1e100.net (142.250.178.14): icmp_seq=3 ttl=128 time=10.9 ms
64 bytes from lhr48s27-in-f14.1e100.net (142.250.178.14): icmp_seq=4 ttl=128 time=10.7 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3010ms
rtt min/avg/max/mdev = 8.257/9.654/10.935/1.169 ms
ubuntuusername@ubuntuservername:~$ _
```

```
ubuntuusername@ubuntuservername:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:36:bc:56 brd ff:ff:ff:ff:ff:ff
    altnme enp2s1
    inet 192.168.80.137/24 metric 100 brd 192.168.80.255 scope global dynamic ens33
        valid_lft 1731sec preferred_lft 1731sec
    inet6 fe80::20c:29ff:fe36:bc56/64 scope link
        valid_lft forever preferred_lft forever
ubuntuusername@ubuntuservername:~$
```

I also changed the network setting so a static IP address would be used instead this is to ensure that the server always keeps the same address which can be very important when running a database or website

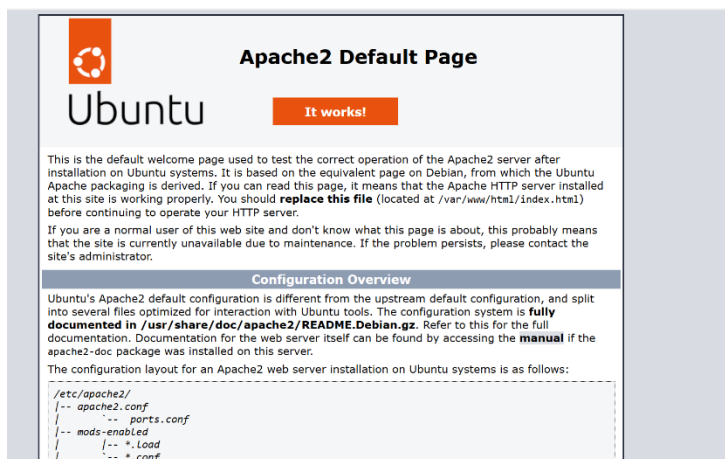
Screenshot



To set php I created a file in notepad called info.php and saved it within the Apache web folder (/var/www/html)

```
ubuntuusername@ubuntuservername:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Mon 2025-05-19 22:08:20 UTC; 4min 22s ago
     Docs: https://httpd.apache.org/docs/2.4/
   Process: 893 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
    Main PID: 960 (apache2)
       Tasks: 7 (limit: 4552)
      Memory: 23.7M (peak: 24.0M)
         CPU: 369ms
    CGroup: /system.slice/apache2.service
           └─ 960 /usr/sbin/apache2 -k start
             968 /usr/sbin/apache2 -k start
             969 /usr/sbin/apache2 -k start
             970 /usr/sbin/apache2 -k start
             971 /usr/sbin/apache2 -k start
             975 /usr/sbin/apache2 -k start
            1555 /usr/sbin/apache2 -k start

May 19 22:08:19 ubuntuservername systemd[1]: Starting apache2.service - The Apache HTTP Server...
May 19 22:08:20 ubuntuservername apachectl[937]: AH00558: apache2: Could not reliably determine the
May 19 22:08:20 ubuntuservername systemd[1]: Started apache2.service - The Apache HTTP Server.
```



PHP App & MySQL Setup

After the installation for MySQL, I ran “sudo mysql_secure_installation” and was able to set up a password. Remove anonymous users, remove root login, and delete test database. By doing these tasks my database becomes more secure from attacks or misconfiguration.

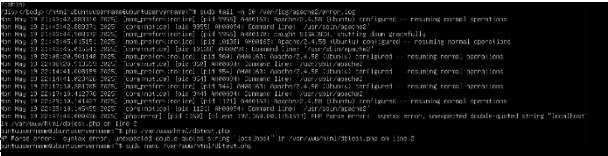
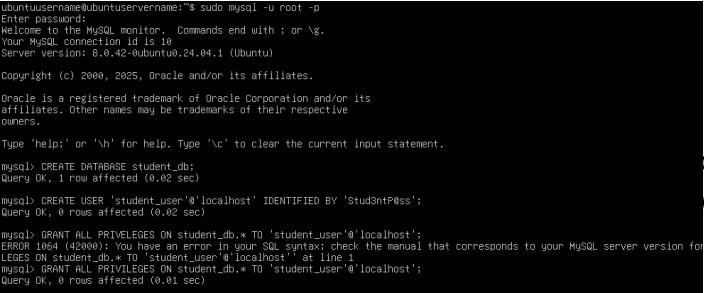
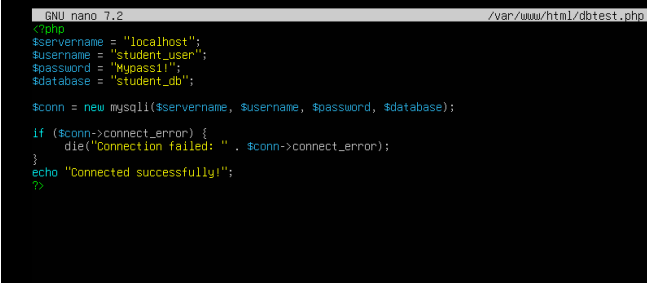
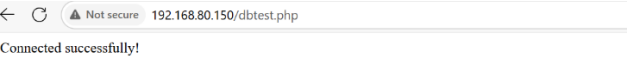
To log in to MySQL I ran “sudo mysql-u root -p” next created a new database and new user with the right authorisations. By giving right permission to the right user security is improved as I don’t have to use the root account access the application

```
ubuntuusername@ubuntuservername:~$ sudo adduser secure
Info: Adding user 'secure' ...
Info: Selecting UID/GID from range 1000 to 59999 ...
Info: Adding new group 'secure' (1001) ...
Info: Adding new user 'secure' (1001) with group 'secure (1001)' ...
Info: Creating home directory '/home/secure' ...
Info: Copying files from '/etc/skel' ...
New password:
Retype new password:
Sorry, passwords do not match.
passwd: Authentication token manipulation error
passwd: password unchanged
Try again? (y/N) y
New password:
Retype new password:
passwd: password updated successfully
Changing the user 'secure' information for 'secure'
Enter the new value, or press ENTER for the default
Full Name []:
Room Number []:
Work Phone []:
Home Phone []:
Other []:
Is the information correct? [Y/n] y
Info: Adding new user 'secure' to supplemental / extra groups 'users' ...
Info: Adding user 'secure' to group 'users' ...
```



Next, I created a dbtest.php file which is a php file that will be sued to test the connection with PHP and MySQL

Once I was able to see the success message in my browser. I was sure the LAMP stack was fully working, and the application and database were connected correctly.



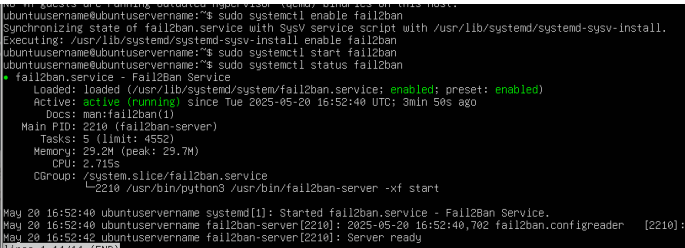
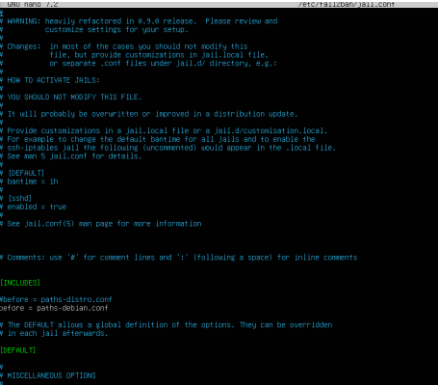
I was able to only allow essential

Apache is used to open ports like 80 and 443 which are used in websites

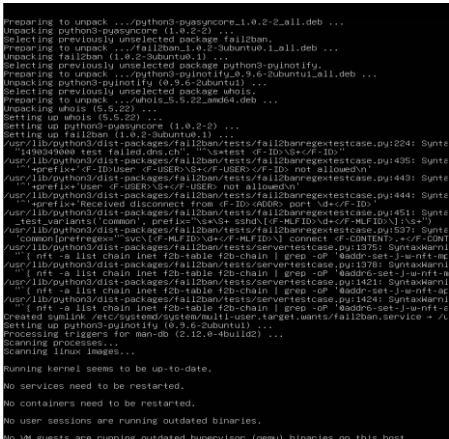
OpenSSH keeps the remote access open through SSH. By blocking other ports, it reduces the chances of an attack

I also added a non-root user as using the root account can be risky and so by using a standard user called secure user and adding sudo privileges I am still able to do tasks that only an admin can do safely and without root

Lastly, I installed Fail2ban as this is known to prevent brute force login attacks, how it does this is by keeping account of login attempts and any ips that keeping failing



ubuntuusername@ubuntuusername:~\$ sudo usermod -aG sudo secure



[illegible]

The screenshot shows two windows from the Oracle VM VirtualBox software. The 'New Virtual Machine Wizard' window is on the left, and the 'Virtual Machine Settings' window is on the right.

New Virtual Machine Wizard:

- Ready to Create Virtual Machine:** The wizard is at the final step. It instructs the user to 'Click Finish to create the virtual machine and start installing Ubuntu 64-bit.'
- Summary:** The virtual machine will be created with the following settings:
 - Name: UbuntuServer11
 - Location: C:\Users\suboo\OneDrive\Documents\Virtual Machines\UbuntuServer11
 - Version: Workstation 17.5 or later
 - Operating System: Ubuntu 64-bit
 - Hard Disk: 18 GB, Split
 - Memory: 4096 MB
 - Network Adapter: NAT
 - Other Devices: 2 CPU cores, CD/DVD, USB Controller, Sound Card
- Buttons:** 'Customize Hardware...' and 'Finish' (highlighted in blue). A checkbox 'Power on this virtual machine after creation' is checked.

Virtual Machine Settings:

- Hardware - Options:** A list of hardware components with a summary table.

Device	Summary
Memory	4 GB
Processors	2
Hard Disk (SCSI)	18 GB
CD/DVD (SATA)	Using file C:\Users\suboo\p...
Network Controller	VirtIO
USB Controller	Present
Sound Card	Auto detect
Display	Auto detect
- Device Status:** A list of devices with their status.

Device	Status
Connected	☒
Connect at power on	☒
- Network connection:** The network adapter is configured as NAT.
 - ☐ Bridged: Connected directly to the physical network
 - ☐ Replicate physical network connection state
 - ☒ NAT: Used to share the host's IP address
 - ☐ Host-only: A private network shared with the host
 - ☐ Custom: Specific virtual network
 - ☐ Internal3 (Auto-bridging)
 - ☐ LAN segment:
- Buttons:** 'LAN Segments...' and 'Advanced...'.

Vulnerability Mitigation & Disaster Recovery

Online Application Vulnerability Mitigation

Identified Threats & Mitigation Strategies

Websites are often very targeted using attacks specifically directed at their vulnerabilities which often lie with insecure code or bad configurations. The main threats I considered were the one I know to be most common.

SQL Injection:

SQL injection can allow attackers to “interfere with queries” [4]. In the case that this attack is not properly handled and mitigated, attackers could not only gain access but also destroy the database completely.

Mitigation:

To prevent SQL injections, I used prepared statements in PHP “`mysqli_prepare()`” this prevents malicious code from being used by attackers.

Cross-Site Scripting (XSS)

XSS is used by an attacker to inject JavaScript into web pages, next it executes in a new browser, usually stealing cookies or saved session tokens.

Mitigation:

All input and output entered by users is sanitized using “`htmlspecialchars()`” this is its own function used in PHP and is used to mitigate against XSS attacks through using special characters in a user's input and not reading it as HTML in the browser.

Password Security and Authentication

Mitigation:

Passwords are hashed by using “`password_hash()`”

Linux monitor login attempts and block them when they experience multiple failures.

Keeping Software Updated

By using “`sudo apt update && sudo apt upgrade -y`” on Linux this enables me to be sure that the system is aware of any updates and can install them [5] and on Windows server automatic updates are on and the latest patches have been installed by using Windows update. In addition to installation, I've also installed packages specifically Apache, PHP and XAMPP and these were all installed from secure and official sources. Within Ubuntu configuring UFW made sure to only allow HTTP/HTTPS and SSH.

Using Windows Defender Firewall allowed me to make configuration to block any ports that aren't needed and only needed for DBS, IIS(HTTP), MySQL or RDP.

Backup & Disaster Recovery Strategy

Using the Windows server backup has lots of benefits for example it's built in so no external installation is required, it's easy to configure and can schedule backups. To implement Windows server backup, you access the installation through server manager. Once this is complete,

configuring daily backups of IIS websites, DNS configuration, Shared Files. All daily backups can be saved into a specific virtual drive or USB or even cloud storage. Which means in the case of a disaster and recovery is needed this can happen by restoring the system from backups

With Linux, the tools that can be used are rsync, cron and mysqldump because they are known to be reliable yet lightweight and built into Linux to rsync the following command is ran “rsync -av/var/www/html/ /backups/html/” and the MySQL database can be dumped daily by using the mysqldump command.

Overall, it is important to take care of application threats and server risk by handling both threats the environment becomes more secure and by implementing a strong back up and disaster recovery systems this ensures that even in worst case scenario servers and data can be restored

Critical Analysis & Reflection

When doing the setup and configuration of both Windows and Ubuntu server environments, I did have to face challenges that can be faced in real IT environment by IT professional. One major issue I faced was through the installation process of XAMPP. After installing unfortunately Apache and MySQL started to fail once I started the services to fix this, I looked at the error logs and through thorough research I realised it may have been due to port conflicts or restrictions with permissions. Therefore, I attempted to change the default ports in XAMPP configuration file however this did not solve the problem and so I took the initiative of running the application with administrative privileges.

Within Linux the issue I had to overcome was an error within the PHP to SQL connection by the end of fixing this problem I knew it was due to incorrect variable names and missing semicolons within the dbtest.php file. I also encountered network related issues when configuring a static IP using NetPlan and to fix this I checked the IP address and modified the subnet mask and gateway to match the VM network NAT settings. Another challenge I faced was when trying to enable BitLocker, to solve this I had to find a way to manually turn on the group policy which allowed BitLocker to work without a compatible TPM as that was what the error had mentioned.

Facing these challenges really expanded my technical knowledge and showed me the importance of troubleshooting by using logs and research. The security measures I implemented also showed me how to apply industry standard practices and apply security layers to create a secure environment especially in the case of data loss.

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